

Maritime detection: YOLO11s Fine Tuning

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What is YOLO11s?

- Latest-gen (2024) Ultralytics YOLO model — better accuracy/speed tradeoff than YOLOv8.
- Single-stage detector: predicts boxes + classes in one pass → real-time capable
Architecture: Backbone → Neck → Detection Head "s" (small) variant — best fit for our 7-class dataset.
- Pretrained on COCO — already knows general visual features (edges, shapes, textures).
- Supports easy layer freezing for transfer learning (freeze=N).

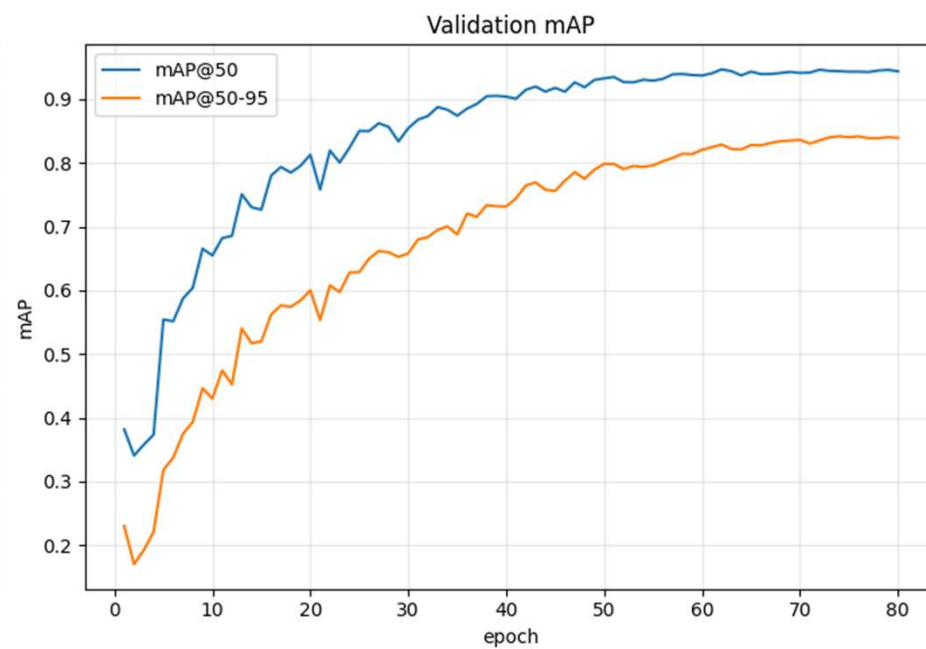
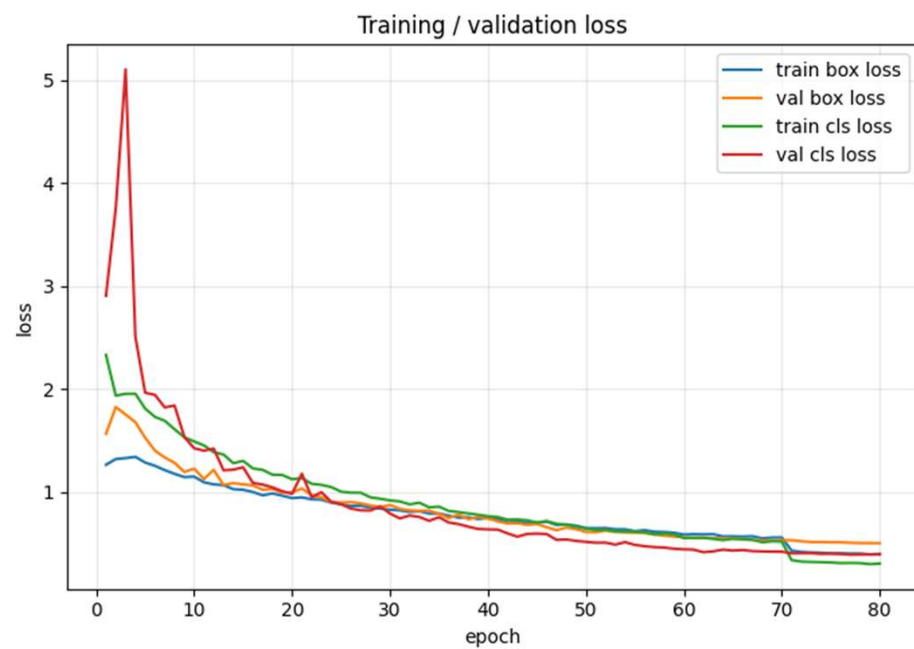
Dataset split after rebalancing

Class	Train (Before → After)	Valid (Before → After)	Test (Before → After)	Total Instances
boat	1,086 → 845	10 → 96	3 → 158	1,099
buoy	396 → 366	43 → 44	17 → 46	456
cargo	378 → 346	36 → 44	17 → 41	431
ferry	498 → 440	37 → 61	19 → 53	554
tug	417 → 379	42 → 44	18 → 54	477
warship	525 → 460	43 → 67	25 → 66	593
yacht	393 → 356	30 → 42	16 → 41	439
ALL CLASSES	3,693 → 3,192	241 → 398	115 → 459	4,049

Fine-Tuning Strategy Used

- Goal:** Adapt COCO-pretrained YOLO11s to 7 marine classes (boat, buoy, cargo, ferry, tug, warship, yacht).
- Strategy:** Partial fine-tuning — freeze early backbone layers (freeze=4), train mid-backbone → neck → head.
- Why:** Marine RGB images are visually close to COCO, so low-level filters (edges/colors) transfer directly — no need for full retraining.
- Config:** 640px, batch 16, 80 epochs, AdamW, lr=0.001, cosine LR schedule.
- Eval:** Best checkpoint tested on held-out set — mAP@50, mAP@50-95, per-class AP.

Important curves



Overall test metrics

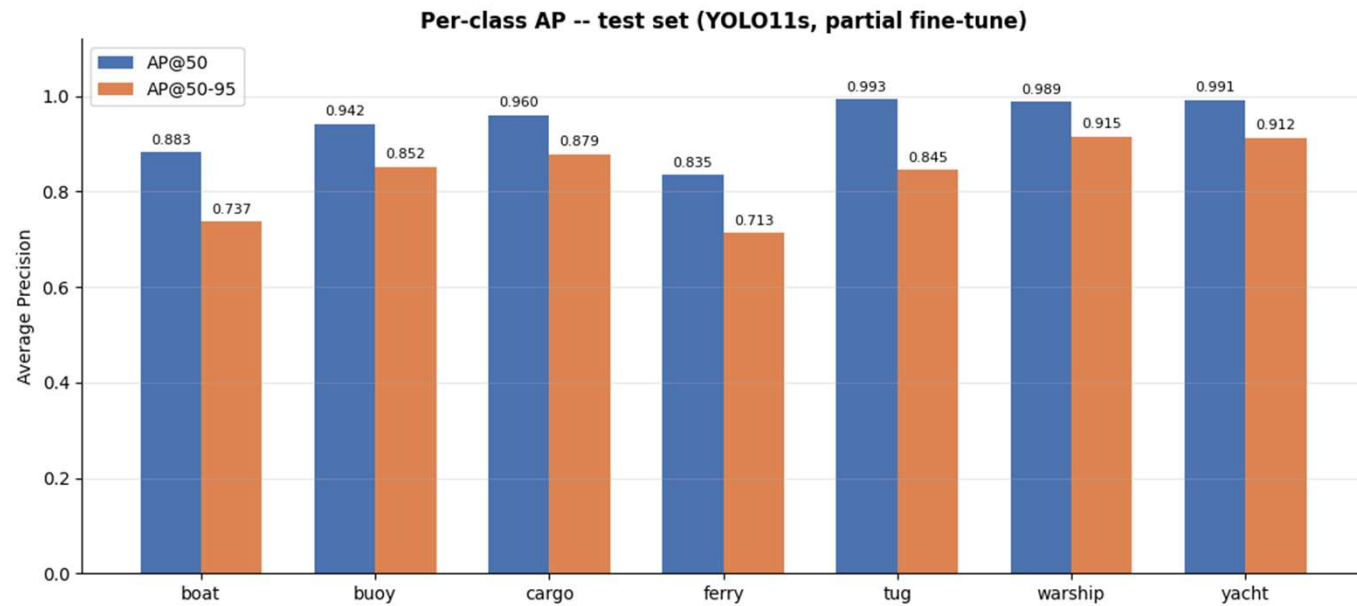
mAP50 : 0.94

mAP50-95 : 0.83

Precision: 0.96

Recall : 0.89

Per class Evaluation metrics



Sample output images: Ground truth vs Predicted

Moored-Kochi-India-Tug-Blue-Sky-Clouds-Stock-Photo-PantherMediaSeller-340039282.jpg.rf.d7f6c2a2534642e7339504d7a7c9d8ca.jpg (1 objects#predicted -- 1 detections



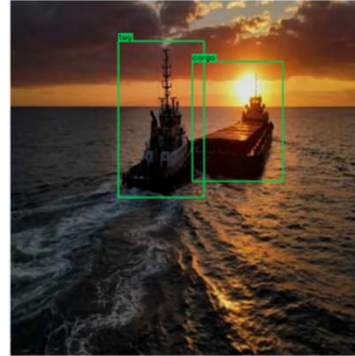
GT -- Cargo-Ships-Anchored-on-the-Bay-Free-Stock-Photo.jpg.rf.b24093e8ab0e95dfa01e62566c964f7.jpg (2 objects)



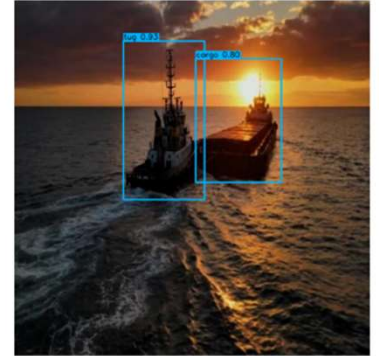
Predicted -- 2 detections



wing-Barge-at-Sunset-Ocean-Logistics-and-Shipping-Operations.jpg.rf.10402c7d3acc2ab68a1af90e6cf4b3b2.jpg (2 objects)



Predicted -- 2 detections



ips-on-Indian-Navy-Day-Parade-with-Tricolor-flag-Stock-Photo-image-of-naval-tricolor_-384805374.jpg.rf.827b3e86db1a90ca1a2901607419edc1.jpg (4 objects)

